

Formative Assessment 4

APM1220 - Applied Multivariate Data Analysis

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Introduction

In this assessment, several multivariate statistical methods will be used to analyze the mean mean vectors, simultaneous confidence intervals, regions, confidence ellipses, and multivariate normality. Additionally, this analysis will also employ the usage of matrix operations, Hotelling's T^2 test, Bonferroni confidence intervals, confidence ellipses, Q-Q plots, and scatter diagrams. Noting this, the medium to be used for this assessment is Rmarkdown, with the results presented and interpreted according to the requirements of each task.

1. Hotelling's T^2 Test for a Mean Vector

Given the data matrix

$$X = \begin{bmatrix} 2 & 12 \\ 8 & 9 \\ 6 & 9 \\ 8 & 10 \end{bmatrix}$$

and the null hypothesis

$$H_0 : \mu' = [7, 11]$$

(a) Evaluate T^2 for testing the null hypothesis $H_0 : \mu' = [7, 11]$

The Hotelling's T^2 statistic is

$$T^2 = n(\bar{\mathbf{x}} - \mu_0)' S^{-1} (\bar{\mathbf{x}} - \mu_0)$$

```
X <- matrix(c(
  2, 12,
  8, 9,
  6, 9,
  8, 10
), ncol = 2, byrow = TRUE)

colnames(X) <- c("X1", "X2")
```

```

n <- nrow(X)
p <- ncol(X)

x_bar <- colMeans(X)
S <- cov(X)

mu0 <- c(7, 11)

T2 <- n * t(x_bar - mu0) %*% solve(S) %*% (x_bar - mu0)

x_bar

```

```

## X1 X2
## 6 10

```

```

S

```

```

##          X1          X2
## X1  8.000000 -3.333333
## X2 -3.333333  2.000000

```

```

T2

```

```

##          [,1]
## [1,] 13.63636

```

(b) Specify the exact distribution of T^2 under the null hypothesis for this situation.

For a sample of size n with p variables,

$$\frac{n-p}{p(n-1)} T^2 \sim F_{p, n-p}$$

```

F_statistic <- ((n - p) / (p * (n - 1))) * as.numeric(T2)

F_statistic

```

```

## [1] 4.545455

```

For this problem, $n = 4$ and $p = 2$. Therefore,

$$\frac{4-2}{2(4-1)} T^2 = \frac{1}{3} T^2 \sim F_{2,2}$$

The corresponding F -statistic is

$$F = 4.5455$$

Thus, the exact distribution under H_0 is

$$\frac{1}{3} T^2 \sim F_{2,2}$$

(c) Using parts (a) and (b), test H_0 at the significance level $\alpha = 0.05$. What statistical conclusion do you reach?

The hypotheses are

$$H_0 : \mu' = [7, 11]$$

and

$$H_1 : \mu' \neq [7, 11]$$

```
alpha <- 0.05

# Critical value and p-value
F_critical <- qf(
  1 - alpha,
  df1 = p,
  df2 = n - p
)

p_value <- 1 - pf(
  F_statistic,
  df1 = p,
  df2 = n - p
)

F_critical
```

```
## [1] 19
```

```
p_value
```

```
## [1] 0.1803279
```

The results are

$$F_{\text{critical}} = 19.0000$$

and

$$p\text{-value} = 0.1803$$

Since

$$F = 4.5455 < 19.0000$$

and

$$p = 0.1803 > 0.05,$$

we fail to reject H_0

Interpretation: At the 5% significance level, there is insufficient evidence to conclude that the population mean vector differs from $[7, 11]'$

2. Simultaneous Confidence Intervals and Regions for Grizzly Bear Body Measurements

```
n <- 61

x_bar <- c(
  Weight = 95.52,
  `Body Length` = 164.38,
  Neck = 55.69,
  Girth = 93.39,
  `Head Length` = 17.98,
  `Head Width` = 31.13
)

S <- matrix(c(
  3266.46, 1343.97, 731.54, 1175.50, 162.68, 238.37,
  1343.97, 721.91, 324.25, 537.35, 80.17, 117.73,
  731.54, 324.25, 179.28, 281.17, 39.15, 56.80,
  1175.50, 537.35, 281.17, 474.98, 63.73, 94.85,
  162.68, 80.17, 39.15, 63.73, 9.95, 13.88,
  238.37, 117.73, 56.80, 94.85, 13.88, 21.26
), nrow = 6, byrow = TRUE)

rownames(S) <- names(x_bar)
colnames(S) <- names(x_bar)

x_bar
```

```
##      Weight Body Length      Neck      Girth Head Length Head Width
##      95.52      164.38      55.69      93.39      17.98      31.13
```

```
S
```

```
##      Weight      Body Length      Neck      Girth Head Length Head Width
## Weight      3266.46      1343.97 731.54 1175.50      162.68      238.37
## Body Length 1343.97      721.91 324.25 537.35      80.17      117.73
## Neck        731.54      324.25 179.28 281.17      39.15      56.80
## Girth       1175.50      537.35 281.17 474.98      63.73      94.85
## Head Length 162.68      80.17 39.15 63.73      9.95      13.88
## Head Width  238.37      117.73 56.80 94.85      13.88      21.26
```

(a) Obtain the large-sample 95% simultaneous confidence intervals for the six population mean body measurements.

The large-sample simultaneous confidence interval is

$$\bar{x}_j \pm \sqrt{\frac{\chi_{p,0.95}^2 s_{jj}}{n}},$$

where $p = 6$

```

p <- 6

chi_critical <- qchisq(0.95, df = p)

SE <- sqrt(diag(S) / n)

margin <- sqrt(chi_critical) * SE

large_sample_CI <- data.frame(
  Variable = names(x_bar),
  Mean = x_bar,
  Lower = x_bar - margin,
  Upper = x_bar + margin
)

kable(
  large_sample_CI,
  digits = 3,
  caption = "Large-Sample 95% Simultaneous Confidence Intervals"
)

```

Table 1: Large-Sample 95% Simultaneous Confidence Intervals

	Variable	Mean	Lower	Upper
Weight	Weight	95.52	69.553	121.487
Body Length	Body Length	164.38	152.173	176.587
Neck	Neck	55.69	49.607	61.773
Girth	Girth	93.39	83.488	103.292
Head Length	Head Length	17.98	16.547	19.413
Head Width	Head Width	31.13	29.035	33.225

(b) Obtain the large-sample 95% simultaneous confidence ellipse for mean weight and mean girth.

The confidence ellipse for mean Weight and mean Girth is based on

$$n(\mu - \bar{\mathbf{x}})'S^{-1}(\mu - \bar{\mathbf{x}}) \leq \chi_{2,0.95}^2$$

```

WG_mean <- x_bar[c("Weight", "Girth")]
WG_S <- S[c("Weight", "Girth"), c("Weight", "Girth")]

chi_critical_2 <- qchisq(0.95, df = 2)

theta <- seq(0, 2 * pi, length.out = 500)

circle <- rbind(
  cos(theta),
  sin(theta)
)

```

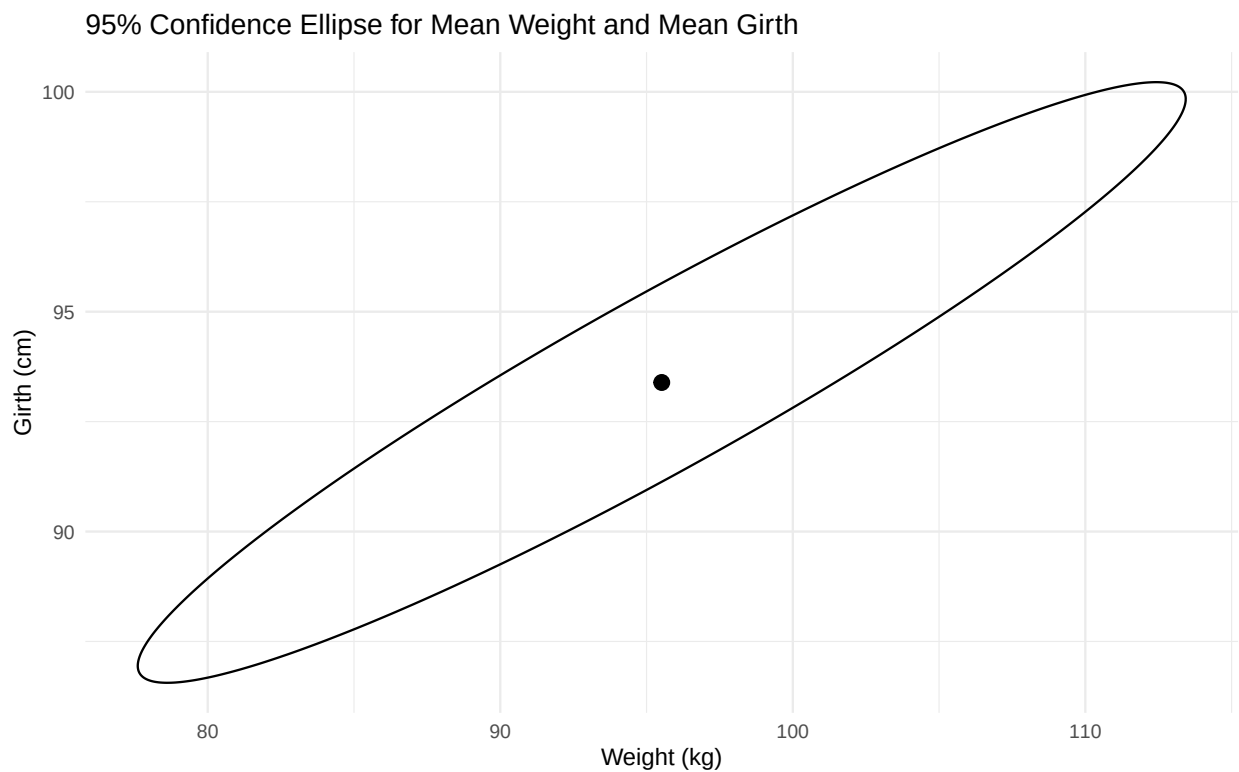
```

ellipse <- t(
  matrix(WG_mean, nrow = 2, ncol = length(theta)) +
    sqrt(chi_critical_2 / n) *
    t(chol(WG_S)) %*% circle
)

ellipse_data <- data.frame(
  Weight = ellipse[, 1],
  Girth = ellipse[, 2]
)

ggplot(ellipse_data, aes(x = Weight, y = Girth)) +
  geom_path() +
  geom_point(
    data = data.frame(
      Weight = WG_mean["Weight"],
      Girth = WG_mean["Girth"]
    ),
    aes(x = Weight, y = Girth),
    size = 3
  ) +
  labs(
    title = "95% Confidence Ellipse for Mean Weight and Mean Girth",
    x = "Weight (kg)",
    y = "Girth (cm)"
  ) +
  theme_minimal()

```



(c) Obtain the 95% Bonferroni confidence intervals for the six means in part (a).

For $m = 6$ simultaneous intervals, the Bonferroni confidence interval is

$$\bar{x}_j \pm t_{1-\alpha/(2m), n-1} \sqrt{\frac{s_{jj}}{n}}$$

```
m <- 6

t_critical <- qt(
  1 - 0.05 / (2 * m),
  df = n - 1
)

bonferroni_margin <- t_critical * SE

bonferroni_CI <- data.frame(
  Variable = names(x_bar),
  Mean = x_bar,
  Lower = x_bar - bonferroni_margin,
  Upper = x_bar + bonferroni_margin
)

kable(
  bonferroni_CI,
  digits = 3,
  caption = "95% Bonferroni Confidence Intervals"
)
```

Table 2: 95% Bonferroni Confidence Intervals

	Variable	Mean	Lower	Upper
Weight	Weight	95.52	75.553	115.487
Body Length	Body Length	164.38	154.993	173.767
Neck	Neck	55.69	51.012	60.368
Girth	Girth	93.39	85.776	101.004
Head Length	Head Length	17.98	16.878	19.082
Head Width	Head Width	31.13	29.519	32.741

(d) Refer to part (b). Construct the 95% Bonferroni confidence rectangle for mean weight and mean girth using $m = 6$. Compare this rectangle with the simultaneous confidence ellipse in part (b).

Using $m = 6$, the Bonferroni intervals for Weight and Girth form the confidence rectangle

```
WG_rectangle <- bonferroni_CI[
  bonferroni_CI$Variable %in% c("Weight", "Girth"),
]

WG_rectangle
```

##	Variable	Mean	Lower	Upper
## Weight	Weight	95.52	75.55331	115.4867
## Girth	Girth	93.39	85.77614	101.0039

The confidence rectangle is defined by

$$75.553 \leq \mu_{\text{Weight}} \leq 115.487$$

and

$$85.776 \leq \mu_{\text{Girth}} \leq 101.004$$

Interpretation: The Bonferroni confidence rectangle is axis-aligned because it is formed from separate confidence intervals for Weight and Girth. In contrast to this, the simultaneous confidence ellipse accounts for the covariance between the two measurements and therefore provides an elliptical joint confidence region.=

(e) Obtain the 95% Bonferroni confidence interval for (mean head width – mean head length) using $m = 6 + 1 = 7$ to allow for this statement as well as statements about each individual mean.

The estimated difference is

$$\bar{x}_{HW} - \bar{x}_{HL}$$

Using $m = 7$, the Bonferroni interval is

$$(\bar{x}_{HW} - \bar{x}_{HL}) \pm t_{1-\alpha/(2m), n-1} \sqrt{\frac{s_{HW,HW} + s_{HL,HL} - 2s_{HW,HL}}{n}}$$

```

difference <- x_bar["Head Width"] - x_bar["Head Length"]

difference_SE <- sqrt(
  (
    S["Head Width", "Head Width"] +
    S["Head Length", "Head Length"] -
    2 * S["Head Width", "Head Length"]
  ) / n
)

m <- 7

t_critical_7 <- qt(
  1 - 0.05 / (2 * m),
  df = n - 1
)

difference_margin <- t_critical_7 * difference_SE

difference_CI <- c(
  Lower = difference - difference_margin,
  Upper = difference + difference_margin
)

```

```
)

difference

## Head Width
##      13.15

difference_SE

## [1] 0.2378179

difference_CI

## Lower.Head Width Upper.Head Width
##      12.48757      13.81243
```

The estimated difference is

$$31.13 - 17.98 = 13.15$$

The resulting 95% Bonferroni confidence interval is

$$(12.488, 13.812)$$

3. Mineral Analysis of Ancient Peruvian Hairs

(a) Construct and plot a 90% joint confidence ellipse for the population mean vector $\mu' = [\mu_1, \mu_2]$, assuming these nine hairs represent a random sample from a particular ancient Peruvian culture.

```
hair <- matrix(c(
  0.48, 12.57,
  40.53, 73.68,
  2.19, 11.13,
  0.55, 20.03,
  0.74, 20.29,
  0.66, 0.78,
  0.93, 4.64,
  0.37, 0.43,
  0.22, 1.08
), ncol = 2, byrow = TRUE)

colnames(hair) <- c("Cr", "St")

n <- nrow(hair)
p <- ncol(hair)
```

```
x_bar <- colMeans(hair)
S <- cov(hair)
```

```
x_bar
```

```
##           Cr           St
##  5.185556 16.070000
```

```
S
```

```
##           Cr           St
## Cr 176.0042 287.2412
## St 287.2412 527.8493
```

The sample mean vector is

$$\bar{\mathbf{x}} = \begin{bmatrix} 5.1856 \\ 16.0700 \end{bmatrix}$$

and the sample covariance matrix is

$$S = \begin{bmatrix} 176.0042 & 287.2412 \\ 287.2412 & 527.8493 \end{bmatrix}$$

For a 90% Hotelling confidence region,

$$n(\mu - \bar{\mathbf{x}})' S^{-1} (\mu - \bar{\mathbf{x}}) \leq \frac{p(n-1)}{n-p} F_{p, n-p}(0.90)$$

```
F_critical <- qf(0.90, df1 = p, df2 = n - p)
T2_critical <- (p * (n - 1) / (n - p)) * F_critical
F_critical
```

```
## [1] 3.257442
```

```
T2_critical
```

```
## [1] 7.445582
```

The critical value is

$$F_{2,7}(0.90) = 3.2574$$

and therefore

$$T_{\text{critical}}^2 = 7.4456$$

```

theta <- seq(0, 2 * pi, length.out = 500)

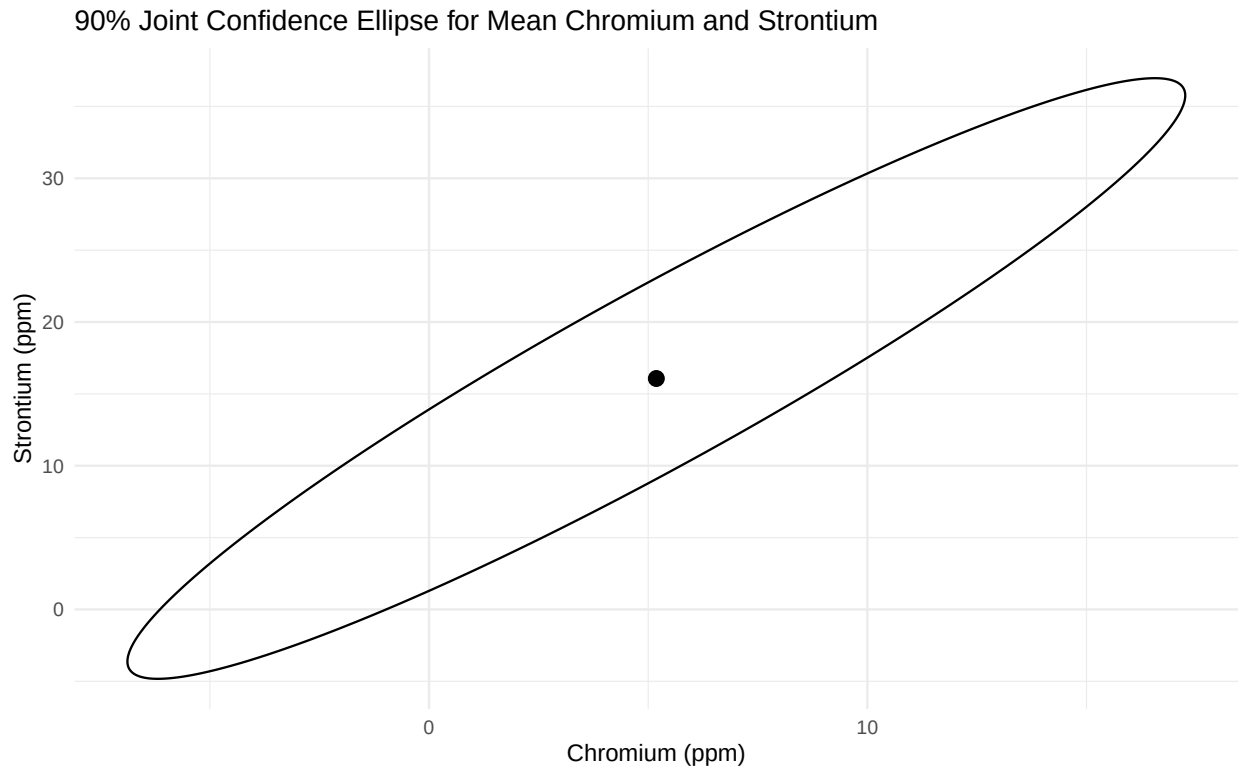
eigen_S <- eigen(S)

ellipse <- t(
  matrix(x_bar, nrow = 2, ncol = length(theta)) +
    sqrt(T2_critical / n) *
    eigen_S$vectors %*%
    diag(sqrt(eigen_S$values)) %*%
    rbind(cos(theta), sin(theta))
)

ellipse_data <- data.frame(
  Cr = ellipse[, 1],
  St = ellipse[, 2]
)

ggplot(ellipse_data, aes(x = Cr, y = St)) +
  geom_path() +
  geom_point(
    data = data.frame(
      Cr = x_bar["Cr"],
      St = x_bar["St"]
    ),
    aes(x = Cr, y = St),
    size = 3
  ) +
  labs(
    title = "90% Joint Confidence Ellipse for Mean Chromium and Strontium",
    x = "Chromium (ppm)",
    y = "Strontium (ppm)"
  ) +
  theme_minimal()

```



(b) Obtain the individual simultaneous 90% confidence intervals for μ_1 and μ_2 by projecting the ellipse constructed in part (a) onto each coordinate axis. Does it appear as if this Peruvian culture has a mean strontium level of 10? That is, are any of the points $(\mu_1 \text{ arbitrary}, 10)$ in the confidence region? Is $[\cdot, 10]'$ a plausible value for μ ? Discuss.

The individual simultaneous confidence intervals are obtained by projecting the confidence ellipse onto each coordinate axis:

$$\bar{x}_j \pm \sqrt{\frac{T_{\text{critical}}^2 s_{jj}}{n}}$$

```
SE <- sqrt(diag(S) / n)

margin <- sqrt(T2_critical) * SE

simultaneous_CI <- data.frame(
  Variable = c("Chromium", "Strontium"),
  Mean = x_bar,
  Lower = x_bar - margin,
  Upper = x_bar + margin
)

kable(
  simultaneous_CI,
  digits = 4,
```

```
caption = "90% Simultaneous Confidence Intervals"
)
```

Table 3: 90% Simultaneous Confidence Intervals

	Variable	Mean	Lower	Upper
Cr	Chromium	5.1856	-6.8812	17.2523
St	Strontium	16.0700	-4.8270	36.9670

The projected 90% simultaneous confidence intervals are approximately

$$-6.8812 \leq \mu_{Cr} \leq 17.2523$$

and

$$-4.8270 \leq \mu_{St} \leq 36.9670$$

The proposed mean strontium value is 10

```
10 >= simultaneous_CI$Lower[2] &&
10 <= simultaneous_CI$Upper[2]
```

```
## [1] TRUE
```

The value 10 is within the projected confidence interval for mean strontium.

Interpretation: A population mean strontium concentration of 10 is consistent with the 90% simultaneous confidence region.

To determine whether

$$\begin{bmatrix} 0.30 \\ 10 \end{bmatrix}$$

is plausible as a joint population mean vector, the point is evaluated against the confidence ellipse.

```
proposed_mean <- c(0.30, 10)

T2_proposed <- n *
  t(proposed_mean - x_bar) %*%
  solve(S) %*%
  (proposed_mean - x_bar)

T2_proposed <- as.numeric(T2_proposed)

T2_proposed
```

```
## [1] 1.772481
```

```
T2_proposed <= T2_critical
```

```
## [1] TRUE
```

The calculated value is

$$T^2 = 1.7725,$$

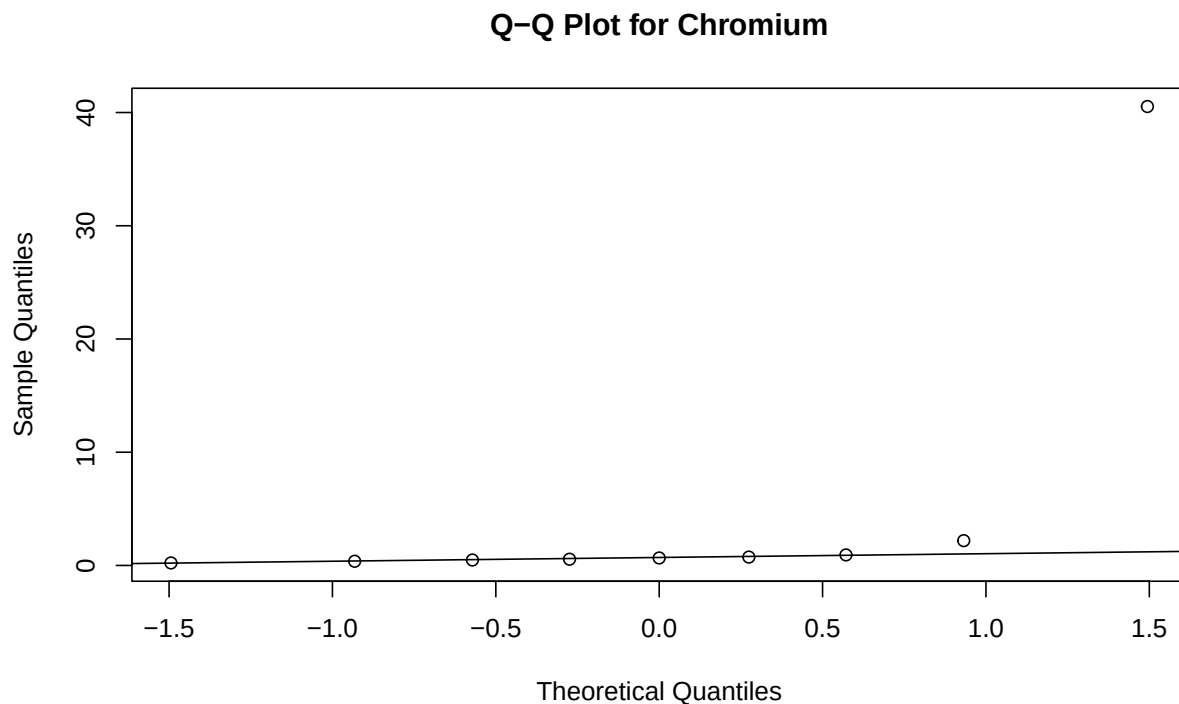
which is less than the critical value

$$7.4456$$

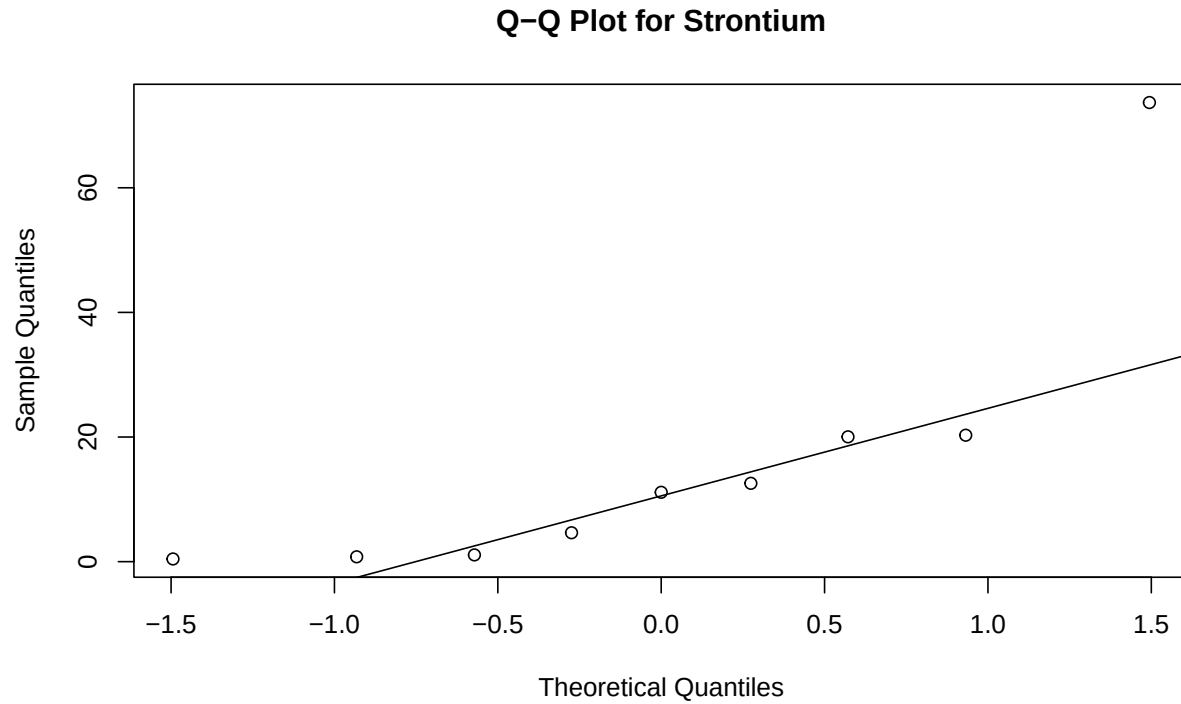
Interpretation: The mean vector $[0.30, 10]'$ lies within the 90% joint confidence ellipse and is therefore plausible based on the sample.

(c) Do these data appear to be bivariate normal? Discuss their status with reference to $Q - Q$ plots and a scatter diagram. If the data are not bivariate normal, what implications does this have for the results in parts (a) and (b)?

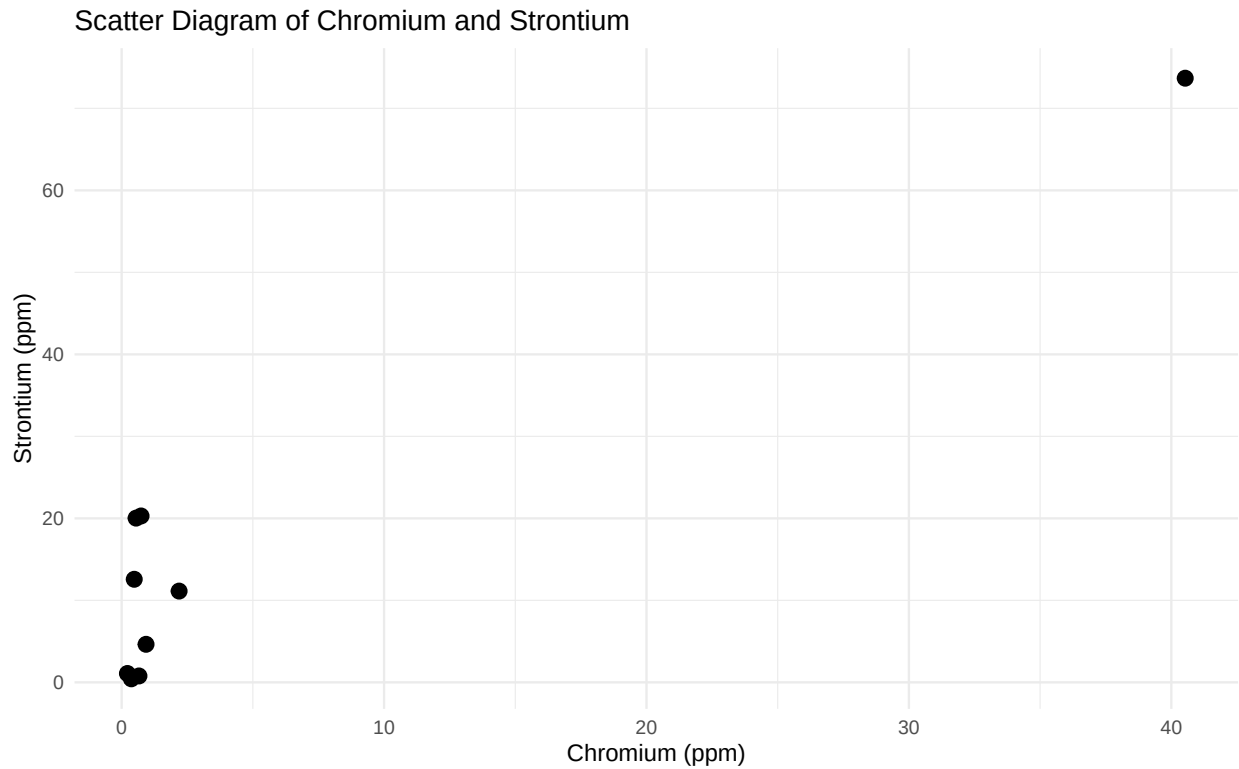
```
qqnorm(hair[, "Cr"],  
       main = "Q-Q Plot for Chromium")  
qqline(hair[, "Cr"])
```



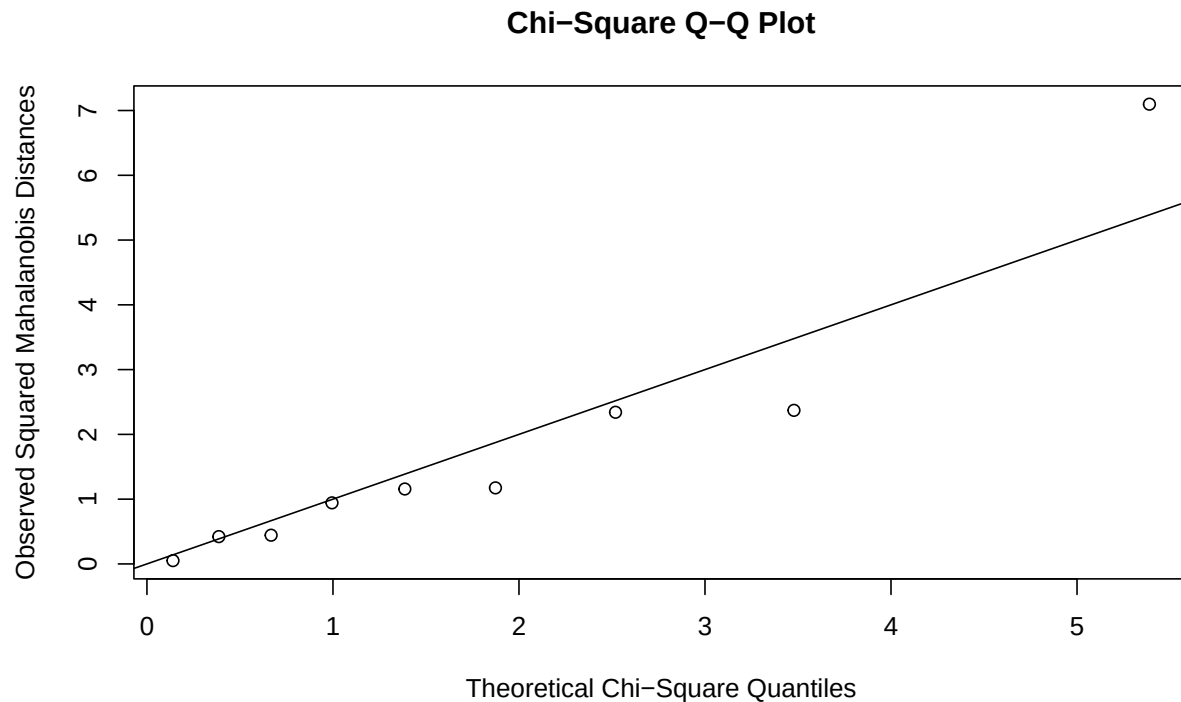
```
qqnorm(hair[, "St"],
       main = "Q-Q Plot for Strontium")
qqline(hair[, "St"])
```



```
ggplot(as.data.frame(hair), aes(x = Cr, y = St)) +
  geom_point(size = 3) +
  labs(
    title = "Scatter Diagram of Chromium and Strontium",
    x = "Chromium (ppm)",
    y = "Strontium (ppm)"
  ) +
  theme_minimal()
```



```
mahalanobis_distances <- mahalanobis(  
  hair,  
  center = x_bar,  
  cov = S  
)  
  
theoretical <- qchisq(  
  ppoints(n),  
  df = p  
)  
  
qqplot(  
  theoretical,  
  sort(mahalanobis_distances),  
  main = "Chi-Square Q-Q Plot",  
  xlab = "Theoretical Chi-Square Quantiles",  
  ylab = "Observed Squared Mahalanobis Distances"  
)  
  
abline(0, 1)
```



Interpretation: From what can be inferred, the data does not appear to be bivariate because the Q-Q plots show noticeable departures from the expected straight-line pattern, while the scatter diagram shows an obvious outlier at (40.53, 73.68). Since bivariate normality is questionable, the 90% confidence ellipse in part (a) and the simultaneous 90% confidence intervals in part (b) may not have their stated coverage probability, so the results of both parts should be interpreted with caution.

(d) Repeat the analysis with the obvious “outlying” observation removed. Do the inferences change? Comment.

```
outlier <- which.max(hair[, "Cr"])
hair_clean <- hair[-outlier, ]
n_clean <- nrow(hair_clean)
x_bar_clean <- colMeans(hair_clean)
S_clean <- cov(hair_clean)
x_bar_clean
```

```
##      Cr      St
## 0.76750 8.86875
```

```
F_critical_clean <- qf(
  0.90,
  df1 = 2,
```

```

    df2 = n_clean - 2
  )

T2_critical_clean <-
  (2 * (n_clean - 1) / (n_clean - 2)) *
  F_critical_clean

F_critical_clean

## [1] 3.463304

T2_critical_clean

## [1] 8.081043

theta <- seq(0, 2 * pi, length.out = 500)

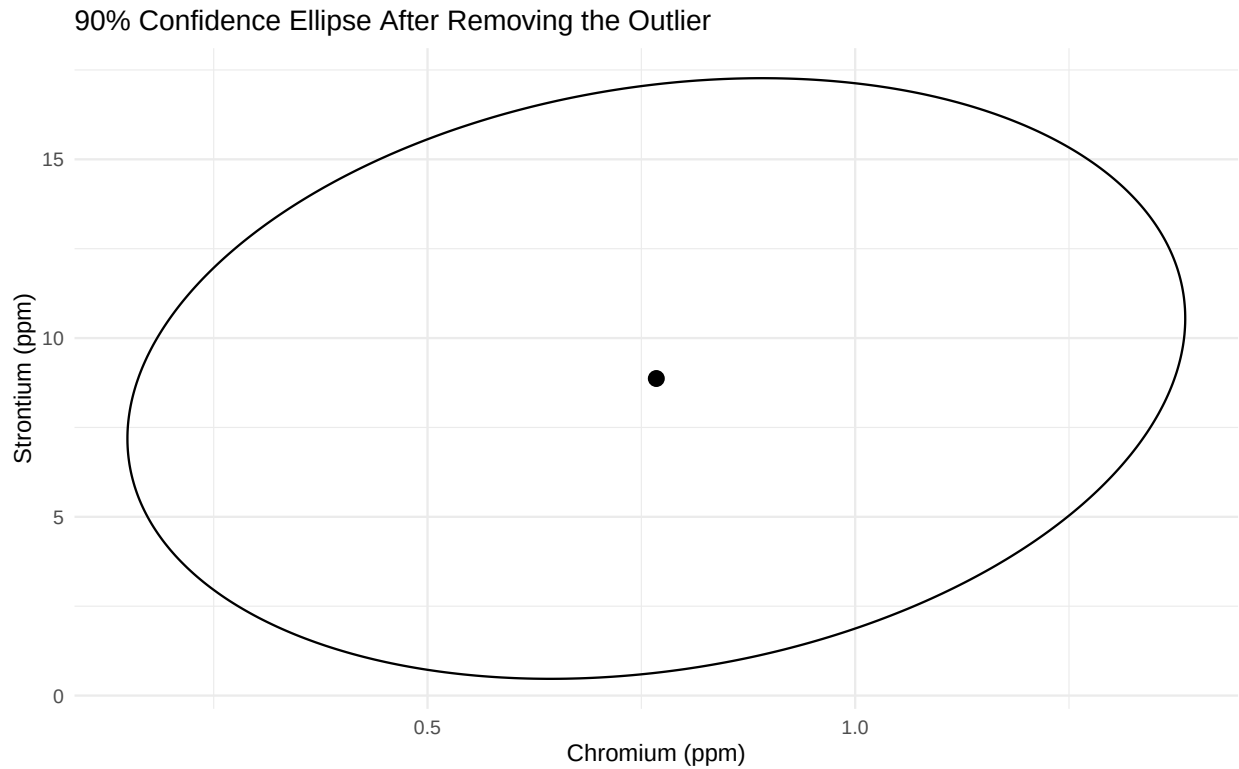
eigen_clean <- eigen(S_clean)

ellipse_clean <- t(
  matrix(
    x_bar_clean,
    nrow = 2,
    ncol = length(theta)
  ) +
  sqrt(T2_critical_clean / n_clean) *
  eigen_clean$vectors %*%
  diag(sqrt(eigen_clean$values)) %*%
  rbind(cos(theta), sin(theta))
)

ellipse_clean_data <- data.frame(
  Cr = ellipse_clean[, 1],
  St = ellipse_clean[, 2]
)

ggplot(ellipse_clean_data, aes(x = Cr, y = St)) +
  geom_path() +
  geom_point(
    data = data.frame(
      Cr = x_bar_clean["Cr"],
      St = x_bar_clean["St"]
    ),
    aes(x = Cr, y = St),
    size = 3
  ) +
  labs(
    title = "90% Confidence Ellipse After Removing the Outlier",
    x = "Chromium (ppm)",
    y = "Strontium (ppm)"
  ) +
  theme_minimal()

```



```
SE_clean <- sqrt(diag(S_clean) / n_clean)

margin_clean <- sqrt(T2_critical_clean) * SE_clean

simultaneous_CI_clean <- data.frame(
  Variable = c("Chromium", "Strontium"),
  Mean = x_bar_clean,
  Lower = x_bar_clean - margin_clean,
  Upper = x_bar_clean + margin_clean
)

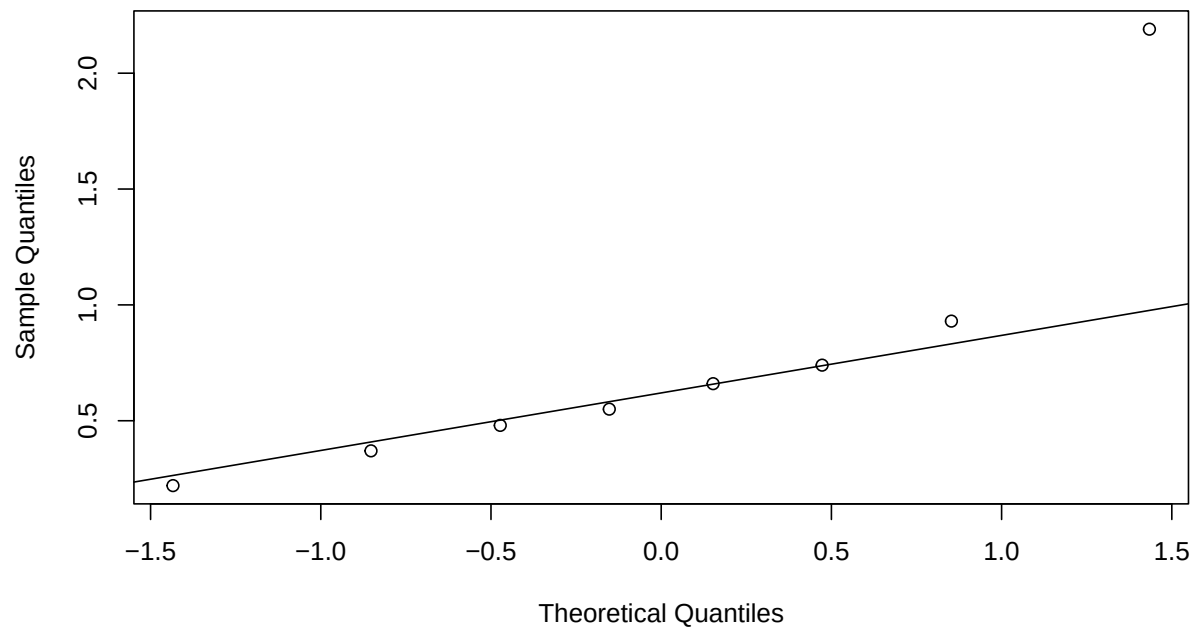
kable(
  simultaneous_CI_clean,
  digits = 4,
  caption = "90% Simultaneous Confidence Intervals After Removing the Outlier"
)
```

Table 4: 90% Simultaneous Confidence Intervals After Removing the Outlier

	Variable	Mean	Lower	Upper
Cr	Chromium	0.7675	0.1491	1.3859
St	Strontium	8.8688	0.4683	17.2692

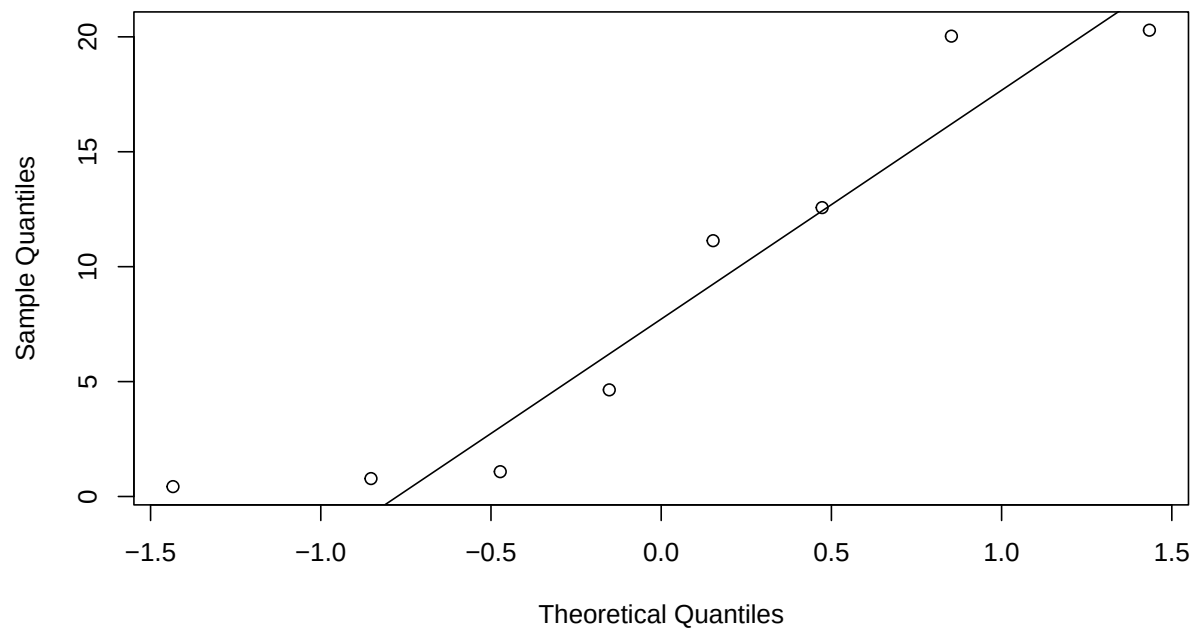
```
qqnorm(hair_clean[, "Cr"],
  main = "Q-Q Plot for Chromium After Removing Outlier")
qqline(hair_clean[, "Cr"])
```

Q-Q Plot for Chromium After Removing Outlier

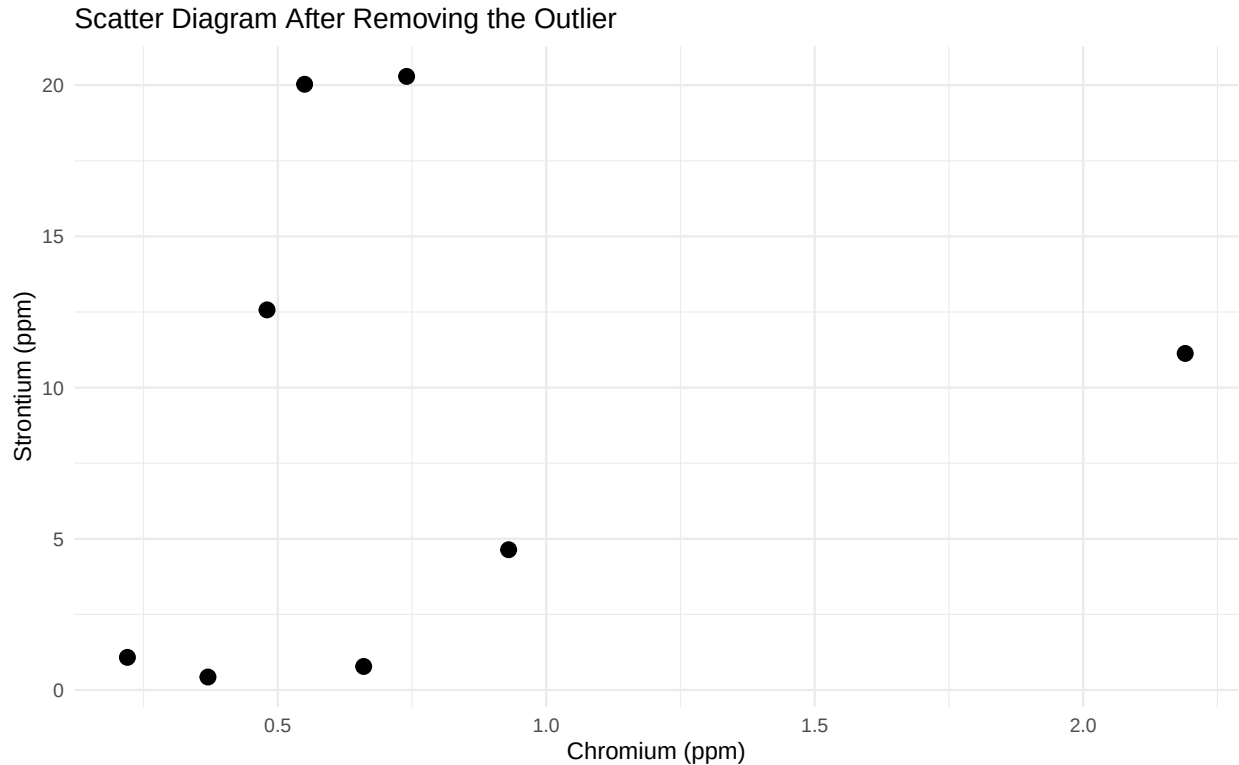


```
qqnorm(hair_clean[, "St"],  
       main = "Q-Q Plot for Strontium After Removing Outlier")  
qqline(hair_clean[, "St"])
```

Q-Q Plot for Strontium After Removing Outlier



```
# Scatter diagram after removing the outlier
ggplot(as.data.frame(hair_clean), aes(x = Cr, y = St)) +
  geom_point(size = 3) +
  labs(
    title = "Scatter Diagram After Removing the Outlier",
    x = "Chromium (ppm)",
    y = "Strontium (ppm)"
  ) +
  theme_minimal()
```



Interpretation: Removing the outlier substantially changes the results. The mean chromium concentration decreases from 5.1856 to 0.7675 ppm, while the mean strontium concentration decreases from 16.0700 to 8.8688 ppm. The confidence region and intervals become much smaller, and the remaining observations show a pattern more consistent with the assumptions of the analysis.

4. Lumber Strength Measurements

(a) Construct and sketch a 95% confidence ellipse for the pair $[\mu_1, \mu_2]'$, where $\mu_1 = E(X_1)$ and $\mu_2 = E(X_2)$.

The sample mean vector is

$$\bar{\mathbf{x}} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}$$

and the 95% confidence ellipse is defined by

$$n(\mu - \bar{\mathbf{x}})'S^{-1}(\mu - \bar{\mathbf{x}}) \leq \frac{p(n-1)}{n-p} F_{p, n-p}(0.95)$$

```
lumber <- matrix(c(
  1232, 4175,
  1712, 7749,
  1115, 6652,
  1932, 6818,
  2205, 7612,
  1820, 9307,
  1897, 10914,
  1900, 6457,
  1932, 10850,
  2426, 10102,
  1612, 7627,
  1558, 7414,
  1598, 6954,
  1470, 7556,
  1804, 8365,
  1858, 7833,
  1752, 9469,
  1587, 8309,
  2067, 6410,
  2208, 9559,
  2365, 10327,
  1487, 6255,
  1646, 7320,
  2206, 10723,
  1579, 8196,
  2332, 5430,
  1880, 9709,
  2540, 12090,
  1773, 10370,
  2322, 10072
), ncol = 2, byrow = TRUE)

colnames(lumber) <- c("Stiffness", "Bending")

n <- nrow(lumber)
p <- ncol(lumber)

x_bar <- colMeans(lumber)
S <- cov(lumber)

x_bar
```

```
## Stiffness    Bending
## 1860.500    8354.133
```

```
S
```

```
##           Stiffness    Bending
```

```
## Stiffness 124054.7 361620.4
## Bending   361620.4 3486333.2
```

The sample mean vector is

$$\bar{\mathbf{x}} = \begin{bmatrix} 1860.50 \\ 8354.13 \end{bmatrix}$$

and the sample covariance matrix is

$$S = \begin{bmatrix} 124054.67 & 361620.45 \\ 361620.45 & 3486333.15 \end{bmatrix}$$

```
F_critical <- qf(
  0.95,
  df1 = p,
  df2 = n - p
)

T2_critical <-
  (p * (n - 1) / (n - p)) * F_critical

F_critical
```

```
## [1] 3.340386
```

```
T2_critical
```

```
## [1] 6.91937
```

The critical value is

$$T_{\text{critical}}^2 = 6.9194$$

```
theta <- seq(0, 2 * pi, length.out = 500)

eigen_S <- eigen(S)

ellipse <- t(
  matrix(
    x_bar,
    nrow = 2,
    ncol = length(theta)
  ) +
  sqrt(T2_critical / n) *
  eigen_S$vectors %*%
  diag(sqrt(eigen_S$values)) %*%
  rbind(cos(theta), sin(theta))
)

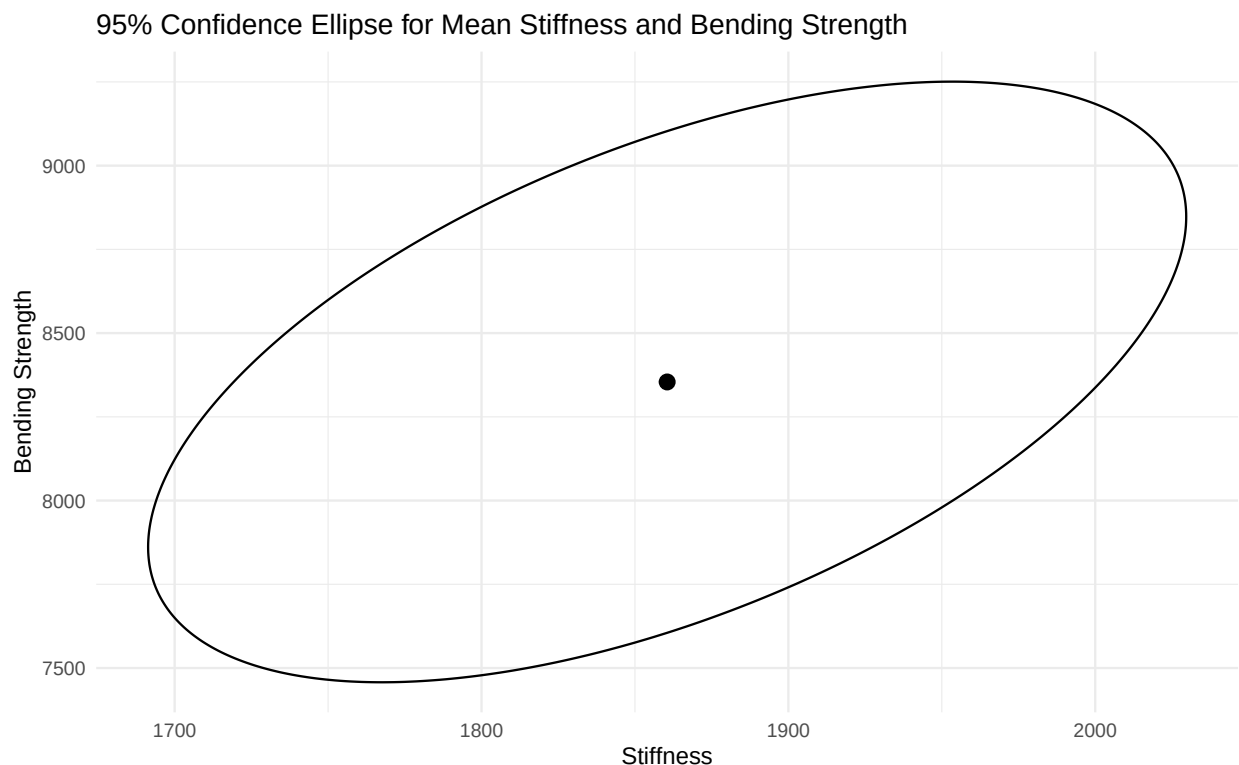
ellipse_data <- data.frame(
```

```

Stiffness = ellipse[, 1],
Bending = ellipse[, 2]
)

ggplot(
  ellipse_data,
  aes(x = Stiffness, y = Bending)
) +
  geom_path() +
  geom_point(
    data = data.frame(
      Stiffness = x_bar["Stiffness"],
      Bending = x_bar["Bending"]
    ),
    aes(x = Stiffness, y = Bending),
    size = 3
  ) +
  labs(
    title = "95% Confidence Ellipse for Mean Stiffness and Bending Strength",
    x = "Stiffness",
    y = "Bending Strength"
  ) +
  theme_minimal()

```



Interpretation: The constructed 95% confidence ellipse is the joint confidence region for $[\mu_1, \mu_2]'$, where $\mu_1 = E(X_1)$ is the population mean stiffness and $\mu_2 = E(X_2)$ is the population mean bending strength. The ellipse is centered at the sample mean vector (1860.50, 8354.13).

(b) Suppose $\mu_{10} = 2000$ and $\mu_{20} = 10,000$ represent “typical” values for stiffness and bending strength, respectively. Given the result in part (a), are the data consistent with these values? Explain.

The proposed typical values are

$$\mu_0 = \begin{bmatrix} 2000 \\ 10000 \end{bmatrix}$$

```
mu0 <- c(2000, 10000)

T2_typical <- n *
  t(mu0 - x_bar) %*%
  solve(S) %*%
  (mu0 - x_bar)

T2_typical <- as.numeric(T2_typical)

F_typical <- ((n - p) / (p * (n - 1))) * T2_typical

p_value <- 1 - pf(
  F_typical,
  df1 = p,
  df2 = n - p
)

T2_typical

## [1] 23.64778

F_typical

## [1] 11.41617

p_value

## [1] 0.0002367286
```

The calculated values are

$$T^2 = 23.6478$$

and

$$p\text{-value} = 0.00024$$

Since

$$23.6478 > 6.9194$$

and

$$p < 0.05,$$

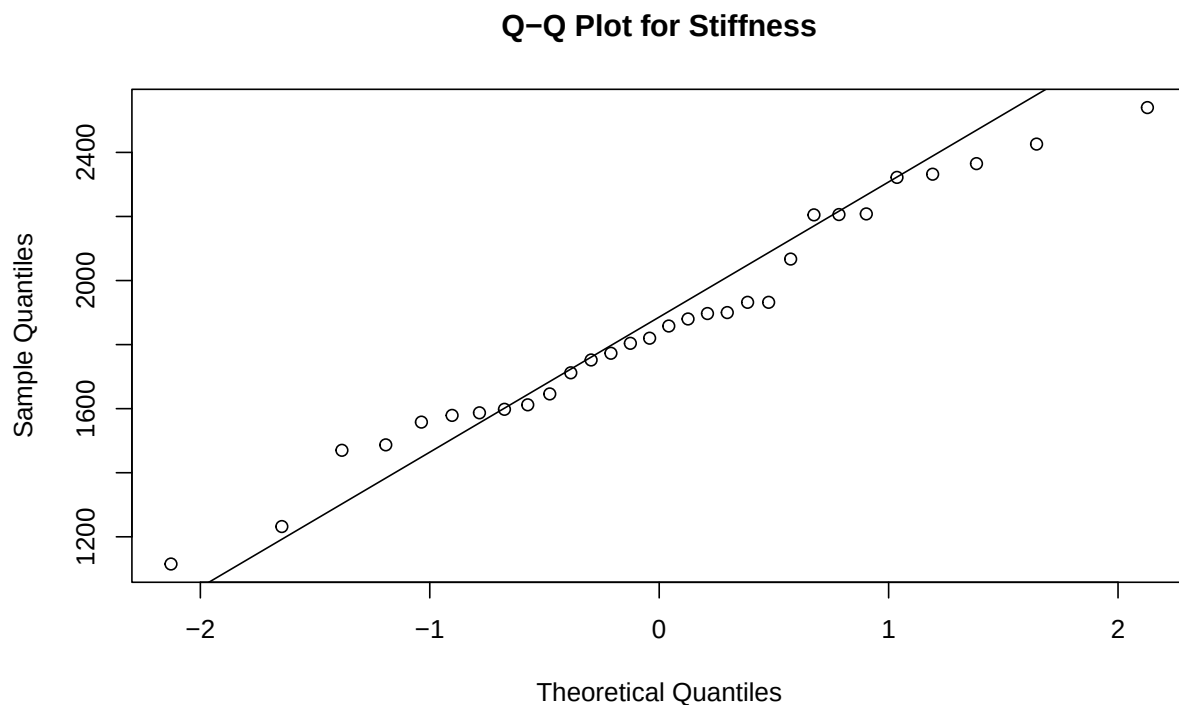
the proposed mean vector lies outside the 95% confidence ellipse

Interpretation: No, the data is not consistent with the typical values $\mu_{10} = 2000$ and $\mu_{20} = 10,000$. The point $(2000, 10,000)$ lies outside the 95% confidence ellipse from part (a), and the corresponding test gives $p = 0.00024 < 0.05$. Therefore, the sample provides evidence that the population mean stiffness and bending strength are not simultaneously equal to 2000 and 10,000, respectively.

(c) Is the bivariate normal distribution a viable population model? Explain with reference to Q – Q plots and a scatter diagram.

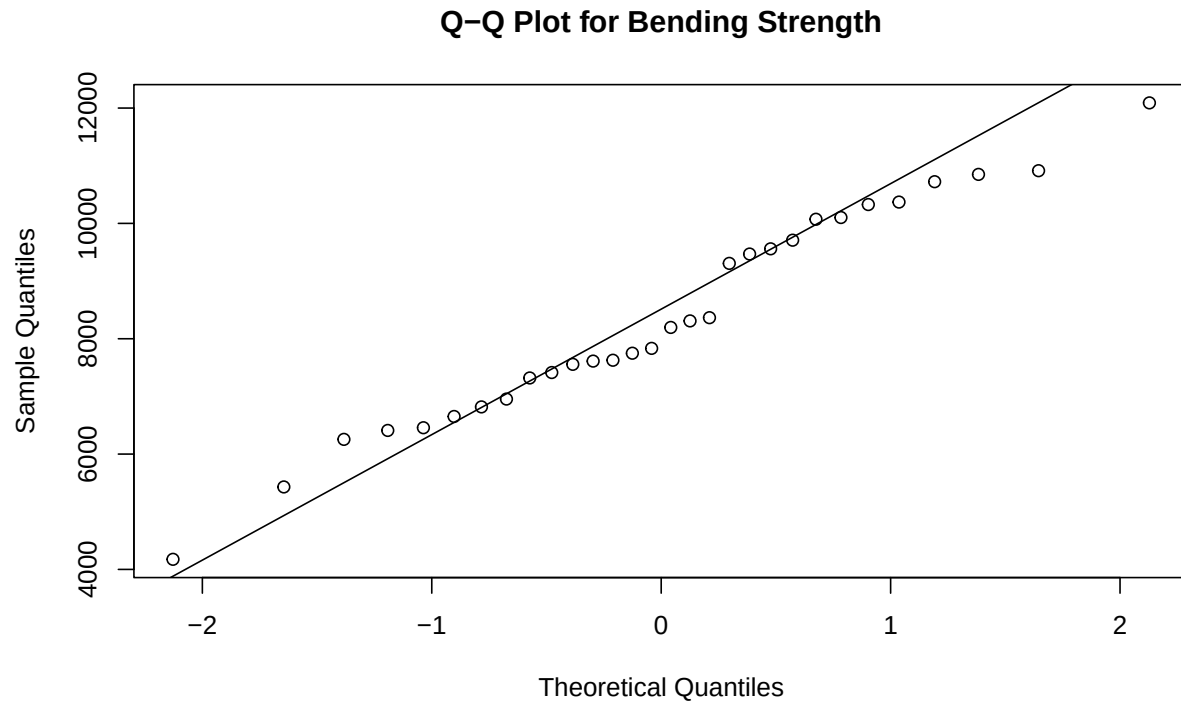
```
# Q-Q plot for Stiffness
qqnorm(
  lumber[, "Stiffness"],
  main = "Q-Q Plot for Stiffness"
)

qqline(lumber[, "Stiffness"])
```

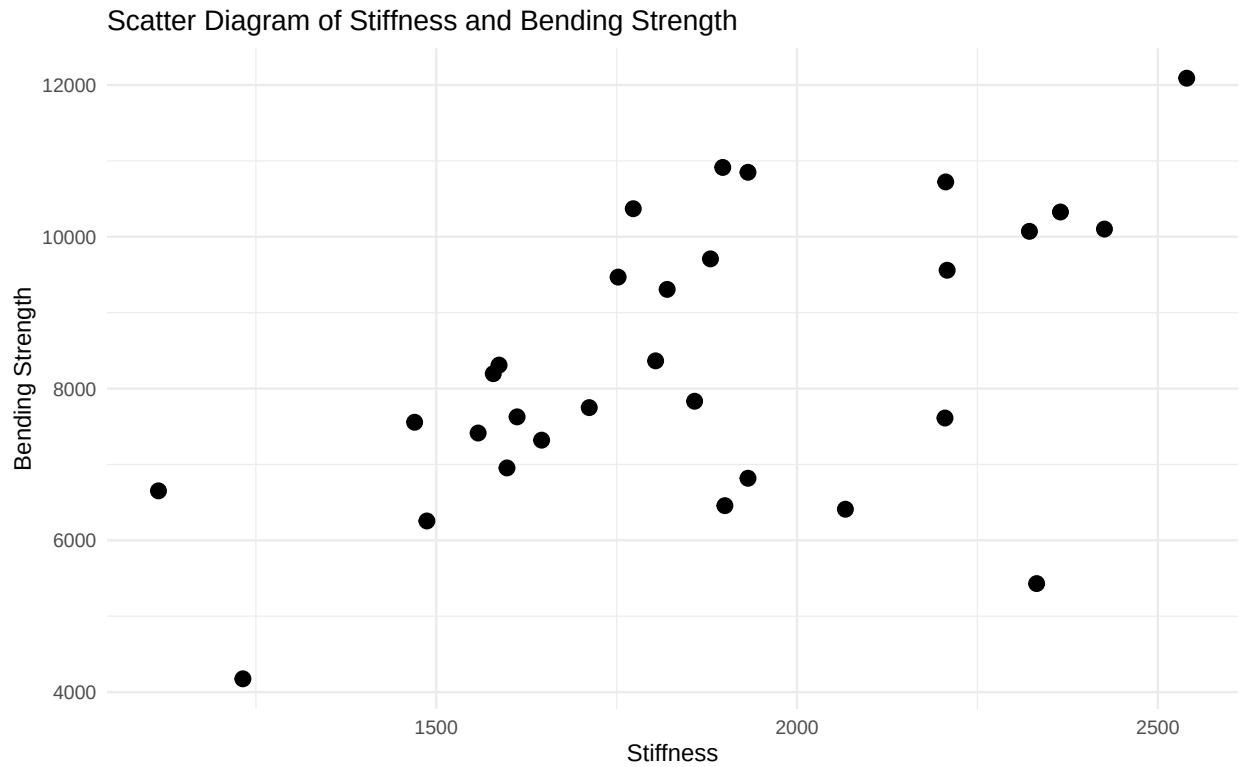


```
# Q-Q plot for Bending Strength
qqnorm(
  lumber[, "Bending"],
  main = "Q-Q Plot for Bending Strength"
```

```
)  
qqline(lumber[, "Bending"])
```



```
# Scatter diagram  
ggplot(  
  as.data.frame(lumber),  
  aes(x = Stiffness, y = Bending)  
) +  
  geom_point(size = 3) +  
  labs(  
    title = "Scatter Diagram of Stiffness and Bending Strength",  
    x = "Stiffness",  
    y = "Bending Strength"  
  ) +  
  theme_minimal()
```



Interpretation: From the distribution above, yes, the bivariate normal distribution appears to be a viable population model for the data. The Q-Q plots show that the observations generally follow the expected straight-line pattern, showing only minor deviations, while the scatter diagram shows a roughly linear positive relationship between stiffness and bending strength and no obvious extreme outliers. Thus, the Q-Q plots and scatter diagram provide reasonable support for assuming bivariate normality.